

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Practical-2

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2. Core JavaScript Concepts for MERN

a. Create functions and call them with parameters.

Code:

a)

```
1 function calculateSalary(name, basicSalary){
2   let bonus = 5000;
3   let totalSalary = basicSalary + bonus;
4
5   console.log("Employee Name:" , name);
6   console.log("Basic salary:" , basicSalary);
7   console.log("Bonus:" , bonus);
8   console.log("Total Salary:" , totalSalary);
9 }
10
11 calculateSalary("Radha", 30000);
12 console.log("radhika");
```

b)

```
function generateBill(product, quantity, price){
  let total = quantity * price;
  let bill = total +(total * 0.09);

  console.log("Product Name:" , product);
  console.log("Quantity: " , quantity);
  console.log("price per item:" , price);
  console.log("Total bill:" , bill);
}
generateBill("Fridge", 1, 50000);
console.log("radhika");
```

Output:

a)

```
PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract2a.js
Employee Name: Radha
Basic salary: 30000
Bonus: 5000
Total Salary: 35000
radhika
```

b)

```
PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract2a.js
Product Name: Fridge
Quantity: 1
price per item: 50000
Total bill: 54500
radhika
```

b. Work with arrays and objects in JavaScript.

Code:

```

2 let fruits = ["Apple", "Banana", "Cherry", "Guava"];
3 let student = {
4   name: "Radha",
5   age: 18,
6   city: "Mumbai"
7 };
8 console.log("Fruits: ", fruits);
9 console.log("Second Fruits: ", fruits[2]);
10
11 console.log("Student Name: ", student.name);
12 console.log("student Age: ", student.age);
13 console.log("student City: ", student.city);
14 console.log("radhika");

```

Output:

```

PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract2b.js
Fruits: [ 'Apple', 'Banana', 'Cherry', 'Guava' ]
Second Fruits:  Cherry
Student Name:  Radha
student Age:  18
student City:  Mumbai
radhika

```

c. Demonstrate use of arrow functions.

Code:

```

1 //addition
2 let add = (a,b) => {
3   return a+b;
4 };
5 //subtraction
6 let sub = (a,b) => {
7   return a-b;
8 };
9 //multiplication
10 let mul = (a,b) => {
11   return a*b;
12 };
13 console.log("Addition: ", add(10,20));
14 console.log("Subtraction: ", sub(10,8));
15 console.log("Multiply: ", mul(10,4));
16 console.log("radhika");

```

Output:

```

PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract2c.js
Addition:  30
Subtraction:  2
Multiply:  40
radhika

```

d. Write a program using array methods (map, filter, reduce).

Code:

```

1 let numbers = [10, 20, 30, 40, 50];
2 //map()
3 let double = numbers.map(num => num*2);
4 //filter
5 let greater25 = numbers.filter(num => num>25);
6 //reduce
7 let total = numbers.reduce((sum , num) => sum + num,0);
8
9 console.log("original array: ", numbers);
10 console.log("map: ", double);
11 console.log("Filter:" ,greater25);
12 console.log("Reduce: ",total);
13 console.log("radhika");

```

Output:

```

PS C:\Users\itilab\Desktop\radha tyit\mern stact> node pract2d.js
original array: [ 10, 20, 30, 40, 50 ]
map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
radhika

```

e. Create a program that manipulates objects and displays output.

Code:

```

let student = {
  name: "Radha",
  rollno: 18,
  marks: 90
};
console.log("original object:");
console.log(student);
//accesing property
console.log("\n Accesing property");
console.log("Name: ", student.name);
console.log("Marks: ", student.marks);
//updating property
student.marks = 95;
console.log("\nAfter updating marks: \n", student);
//adding new property
student.city = "Mumbai";
console.log("\nafter adding : \n ", student);
//delete property
delete student.rollno;
console.log("\nafter deleteing rollno: \n", student);
console.log("radhika");

```

Output:

```

PS C:\Users\itilab\Desktop\radha tyit\mern stact> node pract2e.js
original object:
{ name: 'Radha', rollno: 18, marks: 90 }

  Accesing property
Name: Radha
Marks: 90

After updating marks:
{ name: 'Radha', rollno: 18, marks: 95 }

after adding :
{ name: 'Radha', rollno: 18, marks: 95, city: 'Mumbai' }

after deleteing rollno:
{ name: 'Radha', marks: 95, city: 'Mumbai' }
radhika

```

f. Implement a small JavaScript program combining functions, arrays, and objects.

Code:

```
//array
let students = [
  {
    rollno: 101,
    name: "DH",
    marks : 80
  },
  {
    rollno: 102,
    name: "Radha",
    marks : 80
  },
  {
    rollno: 1013,
    name: "Abhi",
    marks: 90
  }
];
//function to display student details
function display(s)
{
  console.log("-----student details-----");
  for(let stud of s)
  {
    console.log(`Roll no: ${stud.rollno}`);
    console.log(`Nmae: ${stud.name}`);
    console.log(`Marks: ${stud.marks}`);
    console.log("-----");
  }
}
//function call
display(students);
console.log("radhika");
```

Output:

```
PS C:\Users\itlab\Desktop\radha tyit\mern stact> node pract2f.js
-----student details-----
Roll no: 101
Nmae: DH
Marks: 80
-----
Roll no: 102
Nmae: Radha
Marks: 80
-----
Roll no: 1013
Nmae: Abhi
Marks: 90
-----
radhika
```